

Operating system Assignment

1) Explain how the following applications would benefit from multithreading

- a) web browser b) Word processor c) Multicore system d) Operating system kernel

2) consider the following snapshot of a system

Available resources

R1	R2	R3	R4
2	1	0	0

Current allocation

	R1	R2	R3	R4
P1	0	0	1	2
P2	2	0	0	0
P3	0	0	3	4
P4	2	3	5	4
P5	0	3	3	2

Maximum Demand

	R1	R2	R3	R4
P1	0	0	1	2
P2	2	7	5	0
P3	6	6	5	6
P4	4	3	5	6
P5	0	6	5	2

- a) Is the system currently safe or unsafe state? Why?
- b) Is the system currently deadlocked? Why
- c) Which process may become deadlocked?
- d) If process P3 arrives for (0,1,0,0) can the request be safely granted immediately or not?

3) Perform an analysis for the following sequence of disk requests using FIFO, SSTF, SCAN, C-LOOK

27,129,110,186,147,41,10,64,120. Assume that the disk head is initially positioned over track 100 and is moving in the direction of decreasing track number

4) How are devices represented in unix?

5) Discuss RAID Implementation